

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 9: Geometric Twins

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	9 — Geometric Twins
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This is the hardest tier: expect multi-step chains that combine isosceles base-angle reasoning with Pythagoras, congruence criteria layered on top of CPCT, self-correspondence arguments, and symmetry counts woven into angle work. The recurring traps are SSA disguised as SAS, AAA or equal-area/equal-perimeter mistaken for congruence, the wrong base angle in nested isosceles figures, mixing up the order of rotational symmetry with the count of mirror lines, and reading a correspondence's letter order incorrectly. Do not write on this paper — mark answers on your answer sheet.

1. In triangle ABC, $AB = AC = 17$ cm and $BC = 30$ cm. A point P lies on the base BC with $BP = 9$ cm. The length of the segment AP is:

- (A) 8 cm (B) 10 cm
(C) $\sqrt{145}$ cm (D) 12 cm

2. Triangles ABC and DEF have $AB = DE = 9$ cm, $AC = DF = 12$ cm and $\angle B = \angle E = 90^\circ$. A student writes 'RHS'. What is the most precise correct verdict?

- (A) Congruent, but the right angle is at B/E with hypotenuse $AC/DF = 12$ cm and leg $AB/DE = 9$ cm, so RHS does apply (equal right angle, equal hypotenuse, equal leg)
(B) Not congruent, because angle B is between the two given sides making this the ambiguous SSA case
(C) Congruent by SAS, since the right angle is included between AB and AC
(D) Only similar, because one piece given is a hypotenuse and the other a leg

3. In triangle ABC, $AB = AC$. The bisector of angle B meets AC at P, and $BP = BC$. If $\angle A = 36^\circ$, find $\angle ABP$.

- (A) 36° (B) 72°
(C) 18° (D) 24°

4. A figure is mapped onto itself by reflection in a horizontal mirror line and by reflection in a vertical mirror line through the same centre, but by NO other mirror line. What is its order of rotational symmetry, and could it be a regular polygon?

- (A) Order 4, because two perpendicular mirror lines always force 90° rotational symmetry
(B) Order 2 (the two perpendicular reflections compose to a 180° rotation), and it cannot be a regular polygon with more than 4 sides since those have more than 2 mirror lines
(C) Order 1, since reflections never create rotational symmetry
(D) Order 2, and it must be a square

5. Triangle ABC has angle B = angle C, AB = 13 cm, and the foot of the altitude from A to BC is M with BM = 5 cm. The length AM is:

- (A) 8 cm (B) 18 cm
(C) 12 cm (D) $\sqrt{194}$ cm

6. Triangle ABC $\cong$ triangle DEF in that letter order, and separately triangle DEF $\cong$ triangle PQR in that order. Given angle A = 50° and angle Q = 70° , what is angle C?

- (A) 50° (B) 70°
(C) 60° (D) Cannot be determined

7. In triangle ABC, AB = AC and angle A = 100° . The internal bisectors of the two base angles B and C meet at point I. The measure of angle BIC is:

- (A) 140° (B) 120°
(C) 100° (D) 130°

8. Two triangles ABC and DEF satisfy AB = DE = 6 cm, BC = EF = 8 cm and angle A = angle D = 90° . A student says 'SAS using the right angle.' The correct verdict is:

- (A) SAS, because the right angle is the included angle between AB and BC
(B) SSA, so congruence is not guaranteed
(C) RHS: angle A = 90° is opposite BC, so BC (8 cm) is the hypotenuse and AB (6 cm) a leg; equal right angle, equal hypotenuse, equal leg gives congruence
(D) Only similar, since the second triangle's other leg is unknown

9. A regular polygon has rotational symmetry whose smallest positive angle is 24° . How many lines of symmetry does it have?

- (A) 24 (B) 15
(C) 12 (D) 30

10. Triangle ABC $\cong$ triangle CBA for the SAME triangle (note A and C swapped, B fixed). What is forced, and what is the strongest classification?

(A) $BA = BC$ and angle $A =$ angle C , so the triangle is isosceles with apex at B (and it need not be equilateral)

(B) The triangle must be equilateral (C) angle $B = 90^\circ$

(D) $AB = AC$, with apex at A

11. In triangle ABC , $AB = AC = 10$ cm and $BC = 12$ cm. M is the midpoint of BC and N is the midpoint of AB . The length MN is:

(A) 6 cm

(B) 8 cm

(C) 4 cm

(D) 5 cm

12. Triangle ABC has $AB = AC$. D is on AB and E is on AC with $AD = AE$. Lines BD ... rather, segments BE and CD intersect at O . By which congruence is triangle $ABE \cong$ triangle ACD , and what does it give about OB and OC ?

(A) SSS, and it gives $OB = OD$

(B) ASA, and it gives nothing about O

(C) AAA, so only similarity follows and OB , OC are unrelated

(D) SAS ($AB = AC$, angle A common, $AE = AD$) gives $BE = CD$; and by further congruence $OB = OC$, so triangle OBC is isosceles

13. The angles of a triangle are in the ratio $2 : 3 : 4$, and the triangle is then reflected in a line. How many lines of symmetry does the reflected triangle have, and is it congruent to the original?

(A) 1 line of symmetry, and it is only similar to the original

(B) 0 lines of symmetry, and it is only similar because reflection reverses orientation

(C) 0 lines of symmetry, and it is congruent to the original (reflection preserves all lengths and angles)

(D) 3 lines of symmetry, because the angles are all different

14. In triangle ABC , angle $A = 90^\circ$, $AB = AC$, and the altitude from A meets BC at D . If $BC = 8$ cm, the length AD is:

(A) $4\sqrt{2}$ cm

(B) 4 cm

(C) 8 cm

(D) 2 cm

15. Triangle $ABC \cong$ triangle DEF with correspondence $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Given angle $A = (2x + 15)^\circ$, angle $D = (3x - 5)^\circ$ and angle $B = 50^\circ$, the measure of angle F is:

(A) 75°

(B) 55°

(C) 50°

(D) 65°

16. Two triangles have all three pairs of angles equal, and the perimeter of the first is exactly equal to the perimeter of the second. Can we conclude they are congruent?

(A) No — AAA never gives congruence regardless of any extra information

(B) No — equal perimeter could be split among differently shaped triangles

- (C) Yes — equal corresponding angles make them similar, and equal perimeters then force the scale factor to be 1, so corresponding sides are equal and they are congruent
(D) Only if the triangles are equilateral

17. In triangle ABC, $AB = AC$ and angle $B = 2 \cdot \text{angle } A$. The measure of angle C is:

- (A) 36° (B) 45°
(C) 54° (D) 72°

18. A capital letter in standard block form has rotational symmetry of order 2 but no line of symmetry at all. Which letter is it?

- (A) H (B) X
(C) N (D) T

19. In triangle ABC, AD is the median to BC. It is given that $AD = BD = DC$. What is angle BAC?

- (A) 60° (B) 45°
(C) Cannot be determined (D) 90°

20. Triangles ABC and PQR satisfy $BC = QR$, angle $B = \text{angle } Q$ and angle $C = \text{angle } R$. A student writes 'congruent by SAS.' The correct criterion is:

- (A) SAS, because two angles count as two sides plus the included one
(B) SSA, so congruence is not guaranteed (C) AAA, so only similar
(D) ASA — BC is the side included between the two equal angles B and C, so it is ASA, not SAS

21. An isosceles triangle has an exterior angle at one base vertex equal to 110° . If this base vertex is NOT the apex, the apex angle of the triangle is:

- (A) 70° (B) 110°
(C) 40° (D) 20°

22. Triangle ABC has vertices arranged so that $AB = AC = 5$ cm and $BC = 6$ cm. A second triangle DEF has $DE = DF = 5$ cm and an area equal to triangle ABC's area but $EF \neq 6$ cm. Which is true?

- (A) Triangle DEF can exist (an isosceles triangle with two 5 cm sides can match the area with a different base), so equal area does NOT force congruence
(B) Impossible — equal two sides and equal area always force the third side equal
(C) They must be congruent by SAS
(D) DEF must be the mirror image of ABC, hence congruent

23. Triangle ABC $\cong$ triangle DEF and triangle DEF is the mirror image (reflection) of triangle ABC. If the perimeter of ABC is 30 cm and its area is 35 cm^2 , the perimeter and area of DEF are:

- (A) perimeter 30 cm and area 35 cm^2 — reflection preserves both
- (B) perimeter 30 cm but area possibly different, since a flip can change the enclosed region
- (C) perimeter 60 cm and area 70 cm^2 , because two copies now exist
- (D) perimeter 15 cm and area 17.5 cm^2 , because the mirror halves the figure

24. In triangle ABC, $AB = AC$, and BD is the bisector of angle B meeting AC at D. If angle $BDC = 105^\circ$, then angle A equals:

- (A) 40°
- (B) 50°
- (C) 70°
- (D) 80°

25. A figure has rotational symmetry of order 6. The number of distinct positive rotation angles strictly less than 360° that map it onto itself is:

- (A) 6
- (B) 5
- (C) 60
- (D) 12

26. Triangles ABC and DEF have $AB = DE = 7 \text{ cm}$, $AC = DF = 9 \text{ cm}$ and $BC = EF = 7 \text{ cm}$. Are they congruent, and what extra fact follows about each triangle?

- (A) Congruent by SAS, using the 9 cm side as the included one
- (B) Only similar, since no angle is given
- (C) Not congruent, because two sides are equal and the third differs
- (D) Congruent by SSS, and each triangle is isosceles ($AB = BC = 7 \text{ cm}$), so angle A = angle C in each

27. In triangle ABC, angle $B = 90^\circ$, $AB = 9 \text{ cm}$ and $BC = 12 \text{ cm}$. Triangle DEF is congruent to ABC. The length of the longest side of DEF is:

- (A) 12 cm
- (B) 21 cm
- (C) 15 cm
- (D) $\sqrt{63} \text{ cm}$

28. Triangle ABC is equilateral. It is reflected in the perpendicular bisector of BC, then the image is rotated 120° clockwise about the centre. The final triangle, compared with the original equilateral triangle, is:

- (A) Rotated 120° away from the original and so misaligned
- (B) Congruent and occupying exactly the same region — both a reflection in an axis and a 120° rotation map an equilateral triangle onto itself
- (C) A mirror image that can never coincide with the original
- (D) Larger, because two transformations were applied

29. In triangle ABC, $AB = AC$ and angle $A = 40^\circ$. The bisector of angle C meets AB at E. The measure of angle AEC is:

- (A) 70°
- (B) 110°
- (C) 105°
- (D) 100°

30. A quadrilateral has rotational symmetry of order 2 and exactly two lines of symmetry that pass through opposite vertices (the diagonals), not through edge midpoints. The quadrilateral is:

- (A) a rectangle that is not a square (B) a square
(C) an isosceles trapezium (D) a rhombus that is not a square

31. Triangle $ABC \cong$ triangle DEF , with angle $A = 70^\circ$. In triangle DEF , side $EF = DF$. Using the correspondence $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, what is angle B ?

- (A) 55° (B) 70°
(C) 40° (D) Cannot be determined

32. In an isosceles triangle the apex angle is 4 times each base angle. The apex angle measures:

- (A) 90° (B) 100°
(C) 144° (D) 120°

33. Triangles ABC and DEF have $AB = DE$, $BC = EF$ and angle $A =$ angle D . A student claims SAS. The correct verdict is:

- (A) Not guaranteed — the equal angle $A (= D)$ is opposite BC/EF , not the angle included between the two equal sides AB and BC , so this is the ambiguous SSA case
(B) Congruent by SAS, since two sides and an angle are equal
(C) Congruent by ASA, since one angle and two sides are given
(D) Congruent by RHS, treating angle A as a right angle

34. In triangle ABC , $AB = AC = 25$ cm and $BC = 30$ cm. P is the foot of the perpendicular from B to AC . The length BP is:

- (A) 20 cm (B) 15 cm
(C) 7 cm (D) 24 cm

35. A figure looks identical after rotations of 72° , 144° , 216° and 288° , and has 5 lines of symmetry. Adding one more requirement — that it be a single closed polygon with all sides equal — it must be:

- (A) a five-pointed star outline (pentagram) (B) a regular pentagon
(C) a regular decagon (D) any pentagon with equal sides

36. Triangle ABC has $AB = AC$. The bisector AD of angle A meets BC at D , and it is given $DB = 7$ cm. The length DC is:

- (A) 7 cm (B) 14 cm
(C) 3.5 cm (D) Cannot be determined

37. Two triangles satisfy: angle $A =$ angle $D = 40^\circ$, angle $B =$ angle $E = 60^\circ$, and the side opposite the 80° angle is 10 cm in each. The correct criterion proving congruence is:

- (A) SAS, two sides and the included angle
- (B) SSA, so congruence not guaranteed
- (C) AAS (two angles and a non-included side equal — equivalently the third angles match making it ASA)
- (D) AAA, so only similar

38. In triangle ABC, angle A = angle B = angle C. A line ℓ maps the triangle onto itself by reflection. How many DISTINCT such lines ℓ are there, and what is the order of rotational symmetry?

- (A) 3 lines of symmetry and rotational symmetry of order 3
- (B) 1 line of symmetry and order 1
- (C) 3 lines of symmetry and order 6
- (D) 6 lines of symmetry and order 3

39. Triangle ABC $\cong$ triangle DEF. Given that in triangle ABC, angle A : angle B : angle C = 2 : 3 : 7, the largest angle of triangle DEF is:

- (A) 90°
- (B) 105°
- (C) 70°
- (D) 84°

40. In triangle ABC, AB = AC. D is on BC such that $AD \perp BC$, and E is on AD with EB = EC drawn. Which statement is guaranteed?

- (A) D is the midpoint of BC and E lies on the perpendicular bisector of BC, so BD = DC and EB = EC automatically — the configuration is symmetric about line AD
- (B) E must be the midpoint of AD
- (C) $BD \neq DC$ in general
- (D) angle BAC must be 90°

41. A regular polygon and a regular polygon with twice as many sides are compared. If the smaller one has smallest rotation angle 45° , the LARGER one's smallest rotation angle and number of mirror lines are:

- (A) 90° and 4 mirror lines
- (B) 22.5° and 8 mirror lines
- (C) 22.5° and 16 mirror lines
- (D) 45° and 16 mirror lines

42. Triangle ABC $\cong$ triangle BAC for the SAME triangle (A and B swapped, C fixed). Combined with angle C = 50° , the angles of the triangle are:

- (A) $50^\circ, 50^\circ, 80^\circ$
- (B) $65^\circ, 65^\circ, 50^\circ$
- (C) $60^\circ, 60^\circ, 60^\circ$
- (D) $90^\circ, 40^\circ, 50^\circ$

43. In triangle ABC, AB = AC. Squares are imagined on each side, but ignoring those, the internal bisector of angle B and the internal bisector of angle C meet at I. If angle A = 80° , find angle BIC.

- (A) 100°
- (B) 140°
- (C) 130°
- (D) 120°

44. Triangles ABC and DEF have AB = DE = 8 cm, angle A = angle D = 60° , AC = DF = 8 cm. Which is the most complete true statement?

- (A) Congruent by SAS, and each triangle is equilateral (the 60° included angle between two equal 8 cm sides forces the third side to be 8 cm too)
- (B) Congruent by SAS, but the triangles are only isosceles, never equilateral
- (C) Only similar, since the third side is unknown
- (D) Congruent by SSS, since all three sides are given

45. A point P is reflected in line m to P', then P' is reflected in a line n parallel to m (the two lines a distance 3 cm apart) to give P''. The single transformation taking P to P'' is:

- (A) a reflection in a line midway between m and n
- (B) a rotation of 180° about the midpoint of m and n
- (C) a translation of 3 cm
- (D) a translation through 6 cm (twice the distance between the parallel lines), perpendicular to the lines

46. In triangle ABC, angle B = 90° , AB = 6 cm, BC = 8 cm. M is the midpoint of the hypotenuse AC. The length BM is:

- (A) 5 cm
- (B) 10 cm
- (C) 4 cm
- (D) $\sqrt{50}$ cm

47. Triangle ABC $\cong$ triangle DEF, and you are told only that angle A = angle B and DE = 9 cm. Which length is guaranteed equal to 9 cm?

- (A) AB = 9 cm
- (B) BC = 9 cm
- (C) AC = 9 cm
- (D) Nothing can be concluded

48. An isosceles triangle ABC (AB = AC) has its apex angle increased so that the apex equals the SUM of the two base angles. The apex angle is then:

- (A) 60°
- (B) 120°
- (C) 45°
- (D) 90°

49. Triangles ABC and PQR have AB = PQ, AC = PR, and the median from A to BC equals the median from P to QR. A student claims this forces congruence. The correct verdict is:

- (A) Not congruent, because medians are irrelevant to congruence criteria
- (B) Only similar, since no angle is equal
- (C) Congruent — two sides and the median to the third side equal does determine the triangle (the median to the base, with the two sides, fixes the whole figure)
- (D) Congruent by SAS directly, using angle A

50. A figure has both order-3 rotational symmetry and at least one line of symmetry. The minimum total number of lines of symmetry it must have is:

- (A) 3
- (B) 1
- (C) 6
- (D) 2

51. In triangle ABC, AB = AC and angle A = 20° . The bisector of base angle B meets AC at D. The measure of angle BDC is:

- (A) 80° (B) 60°
 (C) 100° (D) 120°

52. Two triangles are congruent. The first is moved onto the second by a single reflection (one flip, no rotation or slide afterward). Which is fully correct?

- (A) Everything is equal AND the orientation is preserved
 (B) Sides equal but angles possibly changed by the flip
 (C) Only the area is preserved; the perimeter may differ
 (D) All corresponding sides and angles are equal, areas and perimeters are equal, but the orientation (handedness) is reversed

53. In triangle ABC, $AB = AC = 15$ cm and $BC = 18$ cm. The altitude from A meets BC at M, and then the perpendicular from M to AB has foot N. The length AM is:

- (A) 9 cm (B) $\sqrt{306}$ cm
 (C) 12 cm (D) 6 cm

54. Triangle ABC $\cong$ triangle DEF, and angle D = $(x + 40)^\circ$, angle E = $(2x)^\circ$, angle F = $(x - 10)^\circ$. Using the correspondence $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, the measure of angle B is:

- (A) 77.5° (B) 75°
 (C) 70° (D) 37.5°

55. A shape is invariant under a 90° rotation but under NO reflection in any line. The smallest order of rotational symmetry it must have, and its number of mirror lines, are:

- (A) order 4 and 4 mirror lines (B) order 4 and 0 mirror lines
 (C) order 2 and 0 mirror lines (D) order 90 and 0 mirror lines

56. In triangle ABC, AD bisects angle A, BD bisects angle B, and they meet at the incentre, with angle ADB = 110° . Then angle C equals:

- (A) 55° (B) 70°
 (C) 20° (D) 40°

57. Two triangles ABC and DEF have $AB = DE = 5$ cm, $BC = EF = 6$ cm, $CA = FD = 5$ cm. They are congruent. A student also claims each is isosceles AND that angle BAC = angle DEF. Evaluate fully.

- (A) Congruent by SSS, isosceles, and angle BAC = angle DEF is correct
 (B) Only similar, and not necessarily isosceles
 (C) Congruent by SAS, and equilateral
 (D) Congruent by SSS, and each is isosceles ($AB = CA = 5$), so angle B = angle C in each; but angle BAC (the apex) need NOT equal angle DEF, so that part of the claim is false

58. An equilateral triangle of side 12 cm and a square are each given. The square has the same perimeter as the triangle. Comparing their orders of rotational symmetry, the square's order minus the triangle's order equals:

- (A) 0 (B) 1
 (C) 2 (D) -1

59. Triangle ABC has $AB = AC = 13$ cm and $BC = 24$ cm. A second triangle DEF has $DE = DF = 13$ cm and BC 's twin EF arranged so that triangle DEF is congruent to ABC. The area of triangle DEF is:

(A) 156 cm^2

(B) 60 cm^2

(C) 120 cm^2

(D) 78 cm^2

60. Consider the three claims: (i) a figure with exactly one line of symmetry has rotational symmetry of order exactly 1; (ii) a figure with rotational symmetry of order 5 must have an odd number of mirror lines (0 or 5); (iii) a figure with two perpendicular mirror lines has order-4 rotational symmetry. Which combination is correct (T/F)?

(A) All three true

(B) (i) False, (ii) True, (iii) True

(C) (i) True, (ii) True, (iii) False

(D) (i) True, (ii) False, (iii) False

Solutions — Mathematics (Class 7), Chapter 9: Geometric Twins — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	C	31	B	41	C	51	B
2	A	12	D	22	A	32	D	42	B	52	D
3	A	13	C	23	A	33	A	43	C	53	C
4	B	14	B	24	D	34	D	44	A	54	B
5	C	15	A	25	B	35	B	45	D	55	B
6	C	16	C	26	D	36	A	46	A	56	D
7	A	17	D	27	C	37	C	47	A	57	D
8	C	18	C	28	B	38	A	48	D	58	B
9	B	19	D	29	C	39	B	49	C	59	B
10	A	20	D	30	D	40	A	50	A	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Drop the altitude from the apex A to the base; because $AB = AC$, its foot M is the midpoint of BC, so $BM = 30/2 = 15$ cm and angle $AMB = 90^\circ$. In right triangle ABM, $AM = \sqrt{AB^2 - BM^2} = \sqrt{17^2 - 15^2} = \sqrt{289 - 225} = \sqrt{64} = 8$ cm. Since P is on BC with $BP = 9$ cm and $BM = 15$ cm, P lies between B and M with $PM = BM - BP = 15 - 9 = 6$ cm, and AM is perpendicular to BC so triangle AMP is right-angled at M. Hence $AP = \sqrt{AM^2 + PM^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ cm. The value 8 cm is only the altitude AM itself and ignores the horizontal offset PM; $\sqrt{145}$ cm comes from using $BP = 9$ directly as the horizontal leg ($\sqrt{8^2 + 9^2}$) instead of the correct $PM = 6$; 12 cm comes from mis-taking the altitude foot as lying 9 cm from B and then mis-adding, and in any case the shortest distance from A to a point on BC is the altitude 8 cm, so AP just past the foot is only slightly more than 8.

2. (A) Angle B = 90° sits opposite side AC, so AC is the hypotenuse (12 cm) and AB is a leg (9 cm). Equal right angle, equal hypotenuse and one equal leg is exactly RHS, which is a valid criterion, so they are congruent. The SSA-objection fails because RHS is precisely the

safe special case of side-side-right-angle. SAS is wrong because the right angle at B is NOT included between AB and AC (AC does not start at B). 'Only similar' ignores that all three pairs of parts are equal in length.

3. (A) Base angles: $\text{angle ABC} = \text{angle ACB} = (180 - 36)/2 = 72^\circ$. BP bisects angle B, so $\text{angle ABP} = \text{angle PBC} = 36^\circ$ each. (Indeed $BP = BC$ is consistent: triangle BPC has angle $\text{PBC} = 36^\circ$ and angle $\text{BCP} = 72^\circ$, so angle $\text{BPC} = 72^\circ$, making $BP = BC$.) angle $\text{ABP} = 36^\circ$. The trap answer 72° is the whole base angle (not the bisected half); 18° halves the apex instead of the base angle; 24° splits 72° into three.

4. (B) Composing two reflections in lines that meet at angle θ gives a rotation through 2θ ; perpendicular lines ($\theta = 90^\circ$) compose to a 180° rotation, forcing at least order 2. It is not forced to order 4 — that would need a third pair of mirror lines. A non-square rectangle is exactly such a figure (2 mirror lines, order 2), so it need NOT be a square, and any regular polygon with 5+ sides has 5+ mirror lines, contradicting 'exactly two'. Order 1 is wrong because the 180° rotation is genuine.

5. (C) angle B = angle C makes $AB = AC = 13$ cm (isosceles), and the altitude from the apex to the base lands at the midpoint, with right angle at M. In right triangle ABM, $AM^2 = AB^2 - BM^2 = 169 - 25 = 144$, so $AM = 12$ cm. The 8 comes from $13 - 5$; 18 from $13 + 5$; $\sqrt{194}$ from wrongly adding $169 + 25$.

6. (C) Chaining the correspondences $A \leftrightarrow D \leftrightarrow P$, $B \leftrightarrow E \leftrightarrow Q$, $C \leftrightarrow F \leftrightarrow R$. So angle A = angle D = angle P = 50° , and angle Q = angle E = angle B = 70° . The angle sum gives angle C = $180 - 50 - 70 = 60^\circ$. Picking 50° or 70° just echoes a given angle for the wrong vertex; 'cannot be determined' ignores that the correspondences pin down both other angles.

7. (A) Base angles are $(180 - 100)/2 = 40^\circ$ each. The bisectors halve them to 20° each, so in triangle BIC the angles at B and C are 20° each and angle $\text{BIC} = 180 - 20 - 20 = 140^\circ$. (Equivalently angle $\text{BIC} = 90 + A/2 = 90 + 50 = 140^\circ$.) 120° uses $90 + 30$; 100° just copies the apex; 130° comes from $90 + 40$.

8. (C) Because the right angle is at A, the side opposite it, BC, is the hypotenuse. We therefore have equal hypotenuse (8 cm), one equal leg (6 cm) and the right angle — that is RHS, which does guarantee congruence. SAS is wrong because the right angle at A is included between AB and AC, not between AB and BC. It is not the unsafe SSA pattern precisely because the equal angle is the right angle (RHS is the legitimate special case). 'Only similar' is wrong since the third side is then forced equal too.

9. (B) A regular n-gon has smallest rotation $360^\circ/n$, so $360/n = 24$ gives $n = 15$. A regular polygon has exactly n lines of symmetry, hence 15. Answer 24 confuses the angle with a count; 12 halves n; 30 doubles it.

10. (A) The self-correspondence $A \leftrightarrow C$, $B \leftrightarrow B$, $C \leftrightarrow A$ says, by CPCT, side BA = side BC and angle A = angle C. That makes the triangle isosceles with the apex at B; nothing forces all

three sides equal, so it need not be equilateral. The apex is at B (the fixed letter), not A, so 'AB = AC' is the wrong pairing. No right angle is implied.

11. (D) M and N are midpoints of two sides, so MN is the midsegment parallel to the third side AC and equal to half of it: $MN = AC/2 = 10/2 = 5$ cm. The distractor $6 = BC/2$ uses the wrong side; 8 is the altitude AM ($\sqrt{100-36}$); 4 is an unrelated guess. (The numbers 10, 12 are there to tempt a Pythagorean detour that is not needed.)

12. (D) Triangles ABE and ACD share angle A; $AB = AC$ and $AE = AD$ are given, so SAS proves them congruent, giving $BE = CD$ and $\angle ABE = \angle ACD$. Since $\angle ABC = \angle ACB$ (base angles) we get $\angle OBC = \angle OCB$, so triangle OBC is isosceles with $OB = OC$. SSS is wrong (no third pair of sides given directly); ASA is wrong because the common angle A is included between the two equal sides, making it SAS not ASA; AAA never yields congruence and the relation $OB = OC$ genuinely holds.

13. (C) The angles are 40° , 60° , 80° — all different, so the triangle is scalene and has 0 lines of symmetry; reflection does not change that. A reflection is a rigid motion: it preserves every length and angle, so the image is congruent to the original (congruence allows a flip). 'Only similar' wrongly treats a flip as a loss of congruence; '3 lines' is the rule for equilateral, not scalene; the orientation reversal does not break congruence.

14. (B) With $\angle A = 90^\circ$ and $AB = AC$, the triangle is an isosceles right triangle; the altitude from the right angle to the hypotenuse lands at its midpoint and equals half the hypotenuse: $AD = BC/2 = 8/2 = 4$ cm. (In a right triangle the median to the hypotenuse equals half it, and here the altitude coincides with that median.) $4\sqrt{2}$ wrongly applies a leg formula; 8 forgets to halve; 2 quarters it.

15. (A) Corresponding angles are equal, so $2x + 15 = 3x - 5$, giving $x = 20$ and $\angle A = \angle D = 55^\circ$. In triangle ABC the angle sum gives $\angle C = 180 - \angle A - \angle B = 180 - 55 - 50 = 75^\circ$. Since angle F corresponds to angle C, $\angle F = 75^\circ$. The trap 55° is angle A/angle C confusion (it is angle A's value, not angle C's); 50° just echoes angle B; 65° comes from a wrong angle sum.

16. (C) Equal angles give similarity with some scale factor k , so each side of the second is k times the matching side of the first; the perimeter scales by the same k . Equal perimeters force $k = 1$, hence equal corresponding sides and congruence. The blanket 'AAA never gives congruence' ignores the extra equal-perimeter condition. The 'differently shaped' objection contradicts the given equal angles (same shape). It holds for all triangles, not just equilateral ones.

17. (D) Since $AB = AC$, $\angle B = \angle C$. Let $\angle A = a$; then $\angle B = \angle C = 2a$. Sum: $a + 2a + 2a = 180$, so $5a = 180$, $a = 36^\circ$, and $\angle C = 2a = 72^\circ$. The trap 36° is angle A (the apex), not angle C; 45° and 54° come from wrong split assumptions.

18. (C) N maps onto itself under a 180° turn (order-2 rotational symmetry) yet has neither a vertical nor horizontal mirror line — its diagonal stroke breaks both. H and X each have BOTH a vertical and horizontal line of symmetry (so they do have mirror lines), and T has only a vertical mirror line with no order-2 rotation. So N is the unique fit.

19. (D) $BD = DC$ means D is the midpoint of BC, and $AD = BD = DC$ means D is equidistant from A, B, C, so D is the circumcentre lying on BC — i.e. BC is a diameter and A is on the circle, making angle BAC the angle in a semicircle $= 90^\circ$. More elementarily: triangles ABD and ACD are isosceles, so angle ABD = angle BAD and angle ACD = angle CAD; their sum $2(\text{angle BAD} + \text{angle CAD}) = 180^\circ$ gives angle BAC = angle BAD + angle CAD $= 90^\circ$. 60° assumes equilateral; 45° halves wrongly; it is fully determined.

20. (D) Two angles and the side BETWEEN them are equal: BC lies between angle B and angle C, the textbook ASA pattern, which proves congruence. SAS requires two SIDES and the included angle, but only one side is given here, so calling it SAS is wrong. It is not SSA (no two sides given) and not mere AAA (the equal side BC fixes the size).

21. (C) The exterior angle at a base vertex is 110° , so the interior base angle there is $180 - 110 = 70^\circ$. In an isosceles triangle the two base angles are equal, so both are 70° , and the apex angle $= 180 - 70 - 70 = 40^\circ$. The trap 70° gives the base angle, not the apex; 110° reuses the exterior angle; 20° doubles-counts incorrectly.

22. (A) Triangle ABC has base 6, half-base 3, height $\sqrt{(25-9)} = 4$, area $= 12 \text{ cm}^2$. An isosceles triangle with equal sides 5 and base b has height $\sqrt{(25 - b^2/4)}$ and area $= (b/2) \cdot \sqrt{(25 - b^2/4)}$. Setting this to 12 gives $b^2(100 - b^2) = 2304$, whose solutions are $b = 6$ AND $b = 8$ (the 'tall' and 'short' twins, since for $b = 8$ the height is 3 and area is again 12). So a non-congruent DEF with $EF = 8$ exists. Equal two sides plus equal area therefore does NOT force the third side; SAS needs an equal included angle (not given); the mirror-image claim is unjustified.

23. (A) Reflection is a rigid motion: it preserves every length (hence the perimeter, 30 cm) and the size of the enclosed region (hence the area, 35 cm^2); only the left/right orientation is reversed. Areas are NOT changed by a flip, the figure is not duplicated, and nothing is halved.

24. (D) Let angle A = a ; the equal base angles are angle ABC = angle ACB $= (180-a)/2$. BD bisects angle B, so angle DBC $= (180-a)/4$. In triangle BDC the angle sum gives angle BDC $= 180 - \text{angle DBC} - \text{angle ACB} = 180 - (180-a)/4 - (180-a)/2 = 180 - 3(180-a)/4$. Setting this to 105° : $3(180-a)/4 = 75$, so $180 - a = 100$, $a = 80^\circ$. The trap 40° is the base-angle value when $a = 100$; 50° and 70° come from dropping the bisector factor.

25. (B) Order 6 means the mapping angles are multiples of $360/6 = 60^\circ$: 60° , 120° , 180° , 240° , 300° and 360° . Counting only angles STRICTLY less than 360° excludes the 360°

(full turn), leaving five: 60, 120, 180, 240, 300. The answer 6 wrongly includes the full turn; 60 is the smallest angle, not a count; 12 doubles.

26. (D) All three pairs of sides match (7, 9, 7 against 7, 9, 7), so SSS proves congruence. Within each triangle $AB = BC = 7$ cm, so each is isosceles and the angles opposite the equal sides — angle C (opposite AB) and angle A (opposite BC) — are equal. SAS is wrong because no angle was provided; 'only similar' ignores that equal sides also fix size; the third sides do match ($7 = 7$).

27. (C) In ABC the hypotenuse (opposite the right angle at B) is $AC = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$ cm, the longest side. Congruent DEF has the same longest side, 15 cm. 12 is just the longer leg; $21 = 9 + 12$; $\sqrt{63}$ wrongly subtracts.

28. (B) Each symmetry of an equilateral triangle (its 3 mirror lines and its order-3 rotations) maps the triangle onto itself. The reflection in the perpendicular bisector of BC is one of those mirror lines, and a 120° rotation is one of its rotational symmetries; composing two self-maps is still a self-map, so the final figure coincides with the original region. It is not misaligned, not an un-matchable mirror image, and transformations never change size.

29. (C) Base angles: angle B = angle C = $(180 - 40)/2 = 70^\circ$. CE bisects angle C, so angle ACE = 35° . In triangle AEC, angle AEC = $180 - \text{angle A} - \text{angle ACE} = 180 - 40 - 35 = 105^\circ$. The trap 70° is the base angle itself; 110° comes from $180 - 70$; 100° from $180 - 40 - 40$.

30. (D) A rhombus's two lines of symmetry are its diagonals (vertex-to-vertex), and it has order-2 rotational symmetry — matching exactly. A non-square rectangle's two mirror lines pass through the MIDPOINTS of opposite edges, not the vertices, so it fails the 'through vertices' condition. A square has FOUR mirror lines (and order 4). An isosceles trapezium has only ONE line of symmetry and order 1.

31. (B) In triangle DEF, EF is opposite vertex D and DF is opposite vertex E. $EF = DF$ therefore means the opposite angles are equal: angle D = angle E. Since angle D corresponds to angle A = 70° , we get angle D = 70° , so angle E = 70° as well. angle B corresponds to angle E, hence angle B = 70° . The trap 55° comes from treating DE (not EF = DF) as the equal pair and splitting the remainder; 40° is the apex angle F = $180 - 70 - 70$; 'cannot be determined' ignores the isosceles deduction.

32. (D) Let each base angle be b; apex = $4b$. Sum: $b + b + 4b = 180$, so $6b = 180$, $b = 30^\circ$, and apex = $4b = 120^\circ$. The trap 90° comes from $4b = 4 \dots$ mis-solved; 100° and 144° from wrong totals.

33. (A) For SAS the equal angle must lie BETWEEN the two equal sides. Here the two equal sides are AB and BC, whose included angle is angle B, but the given equal angle is angle A — opposite side BC. That is the side-side-angle (SSA) pattern, which does not

guarantee congruence (two non-congruent triangles can satisfy it). So it is not SAS, not ASA (only one angle given), and angle A is not stated to be a right angle so RHS does not apply.

34. (D) First the area: drop the altitude from A to BC; half-base = 15, height = $\sqrt{(25^2 - 15^2)} = \sqrt{(625 - 225)} = \sqrt{400} = 20$, so area = $(1/2)(30)(20) = 300 \text{ cm}^2$. Now BP is the altitude onto AC, so area = $(1/2)(AC)(BP) = (1/2)(25)(BP) = 300$, giving $BP = 600/25 = 24 \text{ cm}$. The 20 is the altitude from A (different altitude); 15 is half the base; 7 is unrelated.

35. (B) Order-5 rotational symmetry plus 5 mirror lines is the symmetry of a regular pentagon; requiring a single closed polygon with all sides equal pins it to the regular pentagon. A pentagram is not a simple (non-self-intersecting) closed polygon. A regular decagon would have order 10 and 10 mirror lines, too many. 'Any equilateral pentagon' is wrong because equal sides alone do not force equal angles (an equilateral pentagon can be irregular and lack the 5-fold symmetry).

36. (A) In isosceles triangle ABC with $AB = AC$, the bisector of the apex angle is also the median to the base, so it bisects BC: $DB = DC$. Hence $DC = 7 \text{ cm}$. 14 doubles; 3.5 halves; it is fully determined by the isosceles symmetry (triangles ABD and ACD are congruent by SAS).

37. (C) Third angle in each is $180 - 40 - 60 = 80^\circ$. The equal side (10 cm) lies opposite the 80° angle, which is angle C (= F). With two equal angles and a corresponding equal side that is not between them, this is AAS — a valid congruence criterion (once two angles match, the third does too, so it reduces to ASA). SAS needs two equal SIDES (only one is given). It is not the unsafe SSA, and not mere AAA because a real side length is fixed.

38. (A) angle A = angle B = angle C forces an equilateral triangle, which has exactly 3 lines of symmetry (each through a vertex and the opposite side's midpoint) and rotational symmetry of order 3 (120° and 240° turns). '1 and 1' describes a scalene/generic triangle; order 6 and 6 mirror lines overcount (that is a regular hexagon's pattern, not a triangle's).

39. (B) Parts sum to $2 + 3 + 7 = 12$, so one part = $180/12 = 15^\circ$. The largest angle is 7 parts = $7 \times 15 = 105^\circ$ (angle C), and the congruent DEF has the same largest angle. 90° wrongly assumes a right angle; 70° uses $180 \times 7/18$ type errors; 84° comes from a wrong part value.

40. (A) In isosceles triangle ABC with $AB = AC$, the altitude AD from the apex is also the perpendicular bisector of the base BC, so $BD = DC$ and every point on line AD (including E) is equidistant from B and C, giving $EB = EC$ for any E on AD. Hence the figure is symmetric about AD. E need not be the midpoint of AD; $BD = DC$ always holds here; no right angle at A is implied.

41. (C) Smallest rotation 45° gives the smaller polygon $n = 360/45 = 8$ sides (a regular octagon). Twice as many sides means 16 (a regular hexadecagon), whose smallest rotation

is $360/16 = 22.5^\circ$ and which has 16 mirror lines. $90^\circ/4$ halves the side count; ' 22.5° and 8' keeps the small count; ' 45° and 16' fails to update the angle.

42. (B) The self-correspondence $A \leftrightarrow B$, $C \leftrightarrow C$ forces $CA = CB$ and angle $A = \text{angle } B$ (isosceles with apex C). With angle $C = 50^\circ$, the two equal base angles are $(180-50)/2 = 65^\circ$ each, giving $65^\circ, 65^\circ, 50^\circ$. ' $50, 50, 80$ ' wrongly makes C a base angle; equilateral 60s ignore angle $C = 50^\circ$; ' $90, 40, 50$ ' invents a right angle.

43. (C) Use angle $BIC = 90^\circ + (\text{angle } A)/2 = 90 + 40 = 130^\circ$ (a standard result for the incentre). Check directly: base angles $= (180-80)/2 = 50^\circ$ each, halved to 25° , so angle $BIC = 180 - 25 - 25 = 130^\circ$. The trap 100° copies double-the-base; 140° uses $A/2 = 50$; 120° uses $A/2 = 30$.

44. (A) $AB = AC = 8$ cm with included angle $A = 60^\circ$ is SAS, so the triangles are congruent. Moreover a triangle with two equal sides and a 60° included angle has its base angles each $(180-60)/2 = 60^\circ$, so all angles are 60° and the third side is also 8 cm — it is equilateral. SSS is wrong because the third side was not GIVEN (it is deduced); 'only similar' ignores the equal SAS data; 'never equilateral' is false here.

45. (D) Reflecting in two parallel lines a distance d apart is equivalent to a translation of $2d$ in the direction perpendicular to the lines. Here $d = 3$ cm, so the net effect is a translation of 6 cm. Two reflections give an even (orientation-preserving) isometry, so the result is not a single reflection; parallel mirrors give a translation, not a rotation; the distance is $2d = 6$, not 3.

46. (A) $AC = \sqrt{(6^2 + 8^2)} = 10$ cm. The median to the hypotenuse of a right triangle equals half the hypotenuse, so $BM = AC/2 = 10/2 = 5$ cm. (M is equidistant from A, B, C — the circumcentre.) 10 forgets to halve; 4 is unrelated; $\sqrt{50}$ wrongly Pythagoras-combines.

47. (A) By the correspondence $A \leftrightarrow D$, $B \leftrightarrow E$, the side DE corresponds to side AB , so $AB = DE = 9$ cm — this follows purely from the congruence and letter order, regardless of the angle fact. (The angle $A = \text{angle } B$ is extra and tells us the triangle is isosceles but is not needed here.) BC corresponds to EF and AC to DF , neither of which equals DE ; 'nothing' ignores the correspondence.

48. (D) Let each base angle be b ; apex $= 2b$ (the sum of the two base angles). Total: $2b$ (apex) $+ b + b = 180$, so $4b = 180$, $b = 45^\circ$, apex $= 2b = 90^\circ$. The trap 60° assumes equilateral; 120° and 45° come from wrong setups. (This is just the right-angled isosceles triangle.)

49. (C) Knowing AB, AC and the median m_a to BC actually pins the triangle: reflect A over the midpoint M of BC to a point A' ; then $ABA'C$ is a parallelogram, so $AA' = 2 \cdot m_a$ and the triangle ABA' has $AB, AA' = 2m_a$ and $BA' = AC$ fixed — SSS fixes it, hence the original triangle is determined up to congruence. So equal AB, AC and equal median force

congruence. 'Medians irrelevant' is false; 'only similar' ignores the fixed lengths; it is NOT plain SAS because angle A is not given (the median substitutes for it).

50. (A) If a figure has rotational symmetry of order 3 (rotations by 120° and 240°) AND one mirror line, then rotating that mirror line by the 120° symmetries produces two more mirror lines, so there must be at least 3 lines of symmetry in total. Having just 1 is impossible alongside order-3 rotation; 6 would require order-6 rotation; 2 cannot occur with odd-order rotation (mirror lines come in a set matching the rotation order). So the minimum is 3.

51. (B) Base angles: angle B = angle C = $(180-20)/2 = 80^\circ$. BD bisects angle B, so angle DBC = 40° . In triangle BDC the angle sum gives angle BDC = $180 - \text{angle DBC} - \text{angle BCD} = 180 - 40 - 80 = 60^\circ$. The trap 80° just reuses a base angle; $100^\circ = 180 - 80$; 120° forgets to bisect angle B (using 40 with a wrong remainder).

52. (D) A reflection is a rigid (distance-preserving) motion, so every length and angle is preserved — equal sides, equal angles, equal perimeter and equal area. But unlike a rotation or translation, a single reflection REVERSES orientation (left becomes right). So handedness is reversed, not preserved; angles are not changed; both area and perimeter are preserved.

53. (C) Because $AB = AC$, the altitude from the apex A bisects the base, so $BM = 18/2 = 9$ cm and angle $AMB = 90^\circ$. In right triangle ABM, $AM^2 = AB^2 - BM^2 = 15^2 - 9^2 = 225 - 81 = 144$, so $AM = 12$ cm. (The point N is a distractor not needed for AM.) 9 is half the base BM, not AM; $\sqrt{306}$ wrongly adds $225 + 81$; 6 halves AM.

54. (B) The angles of DEF sum to 180: $(x + 40) + 2x + (x - 10) = 180$, so $4x + 30 = 180$, $4x = 150$, $x = 37.5$. Then angle E = $2x = 75^\circ$, and angle B corresponds to angle E, so angle B = 75° . The trap $77.5 = x + 40$; 70 mis-solves; 37.5 reports x itself.

55. (B) A 90° rotation that maps the figure onto itself forces all multiples — 90° , 180° , 270° , 360° — so the rotational order is at least 4. The figure can still have NO mirror lines (e.g. a pinwheel/swastika-like motif), so 0 mirror lines is consistent. 'Order 4 and 4 mirror lines' wrongly assumes a square's full symmetry; order 2 is too small (a 90° turn implies order ≥ 4); 'order 90' confuses the angle with the order.

56. (D) In the triangle ABD (using the half-angles at A and B), angle $ADB = 180 - (A/2 + B/2) = 110$, so $A/2 + B/2 = 70$, giving $A + B = 140$. Then angle C = $180 - (A + B) = 180 - 140 = 40^\circ$. (Equivalently angle $ADB = 90 + C/2$, so $110 = 90 + C/2$, $C = 40^\circ$.) The trap 55 = solving for C/... ; $70 = A/2 + B/2$; 20 halves wrongly.

57. (D) Sides 5, 6, 5 match 5, 6, 5, so SSS gives congruence. In each triangle $AB = CA = 5$, so it is isosceles with apex A, making the base angles angle B = angle C equal. The apex angle is angle BAC = angle A = angle D (corresponding), NOT angle DEF (which corresponds to angle B). So angle BAC = angle EDF, not angle DEF — the student's last

equality is wrong. 'Only similar' and 'not isosceles' are both false; it is SSS not SAS, and not equilateral ($5 \neq 6$).

58. (B) The perimeter detail is a distractor — it does not affect symmetry counts. A square has rotational symmetry of order 4; an equilateral triangle has order 3. The difference is $4 - 3 = 1$. The equal-perimeter fact tempts a side-length computation (square side = $36/4 = 9$ cm) that is irrelevant. 0, 2 and -1 mis-subtract.

59. (B) Congruent triangles have equal area, so compute ABC's area: drop the altitude from A to BC; half-base = 12, height = $\sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5$, area = $(1/2)(24)(5) = 60$ cm². Hence DEF also has area 60 cm². $156 = (1/2)(24)(13)$ wrongly uses a side as the height; 120 doubles; $78 = (1/2)(12)(13)$ mixes pieces.

60. (C) (i) True: a single mirror line cannot be rotated onto another by any nontrivial rotation without creating a second mirror line, so the rotational order is 1 (e.g. an isosceles non-equilateral triangle). (ii) True: mirror lines, if any, come in a set whose count equals the rotational order; order 5 means either 0 mirror lines (pinwheel) or exactly 5 — both odd. (iii) False: two PERPENDICULAR mirror lines compose to a 180° rotation, giving order 2, not order 4 (order 4 would need mirror lines at 45°). So the pattern is T, T, F.